1) **Write a ‘C’ program to calculate area and perimeter of a rectangle.**
Ans -> Formula = Area = Length × Breadth
Perimeter = 2 × (Length + Breadth)

```
#include <stdio.h>

int main()
{
   float length, breadth, area, perimeter;

   printf("Enter length: ");
   scanf("%f", &length);

   printf("Enter breadth: ");
   scanf("%f", &breadth);

   area = length * breadth;
   perimeter = 2 * (length + breadth);

   printf("Area of rectangle = %.2f\n", area);
   printf("Perimeter of rectangle = %.2f\n", perimeter);

   return 0;
}
```

OP-
If length = 10 and breadth = 5

- **Area = 10 × 5 = 50**
- **Perimeter = 2 × (10 + 5) = 30**

2) Write a ‘C’ program to calculate the sum of factors of a number using call by references.

```
#include <stdio.h>

void sumOfFactors(int n, int *sum)
{
   int i;

   *sum = 0;

   for (i = 1; i <= n; i++)
   {
      if (n % i == 0)
      {
         *sum = *sum + i;
      }
```

```
    }
}

int main()
{
    int n, sum;

    printf("Enter a number: ");
    scanf("%d", &n);

    sumOfFactors(n, &sum);

    printf("Sum of factors = %d", sum);

    return 0;
}
```

0/p- Enter a number: 6
Sum of factors = 12

3) Write a 'C' program to find the maximum number from an array of 'n' integers.

```
#include <stdio.h>

int main()
{
    int a[100], n, i, max;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    max = a[0];

    for (i = 1; i < n; i++)
    {
        if (a[i] > max)
        {
            max = a[i];
        }
    }
```

```
    printf("Maximum number = %d", max);

    return 0;
}
```

0/p - Enter number of elements: 5
Enter 5 elements:
10 25 7 40 18

Maximum number = 40